

JKBOSE 2026

CLASS 11TH

RESPIRATION IN PLANTS

LECTURE-03

BOTANY

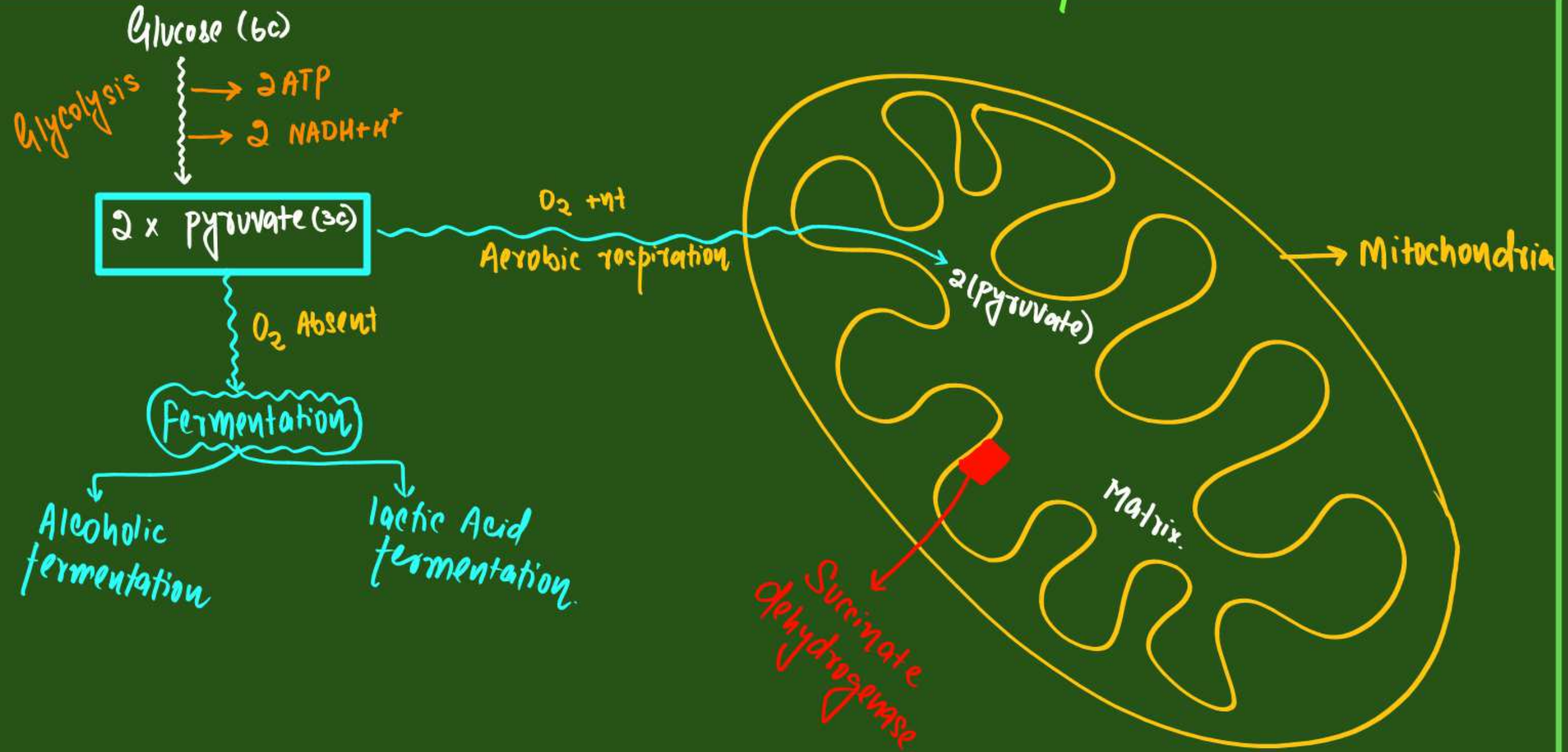
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Topics To Covered

- 1** Aerobic Respiration
- 2** Link Reaction
- 3** Kreb's Cycle
- 4** ATP Balance Sheet

RECAP



AEROBIC RESPIRATION

It occurs inside mitochondria (in presence of oxygen)

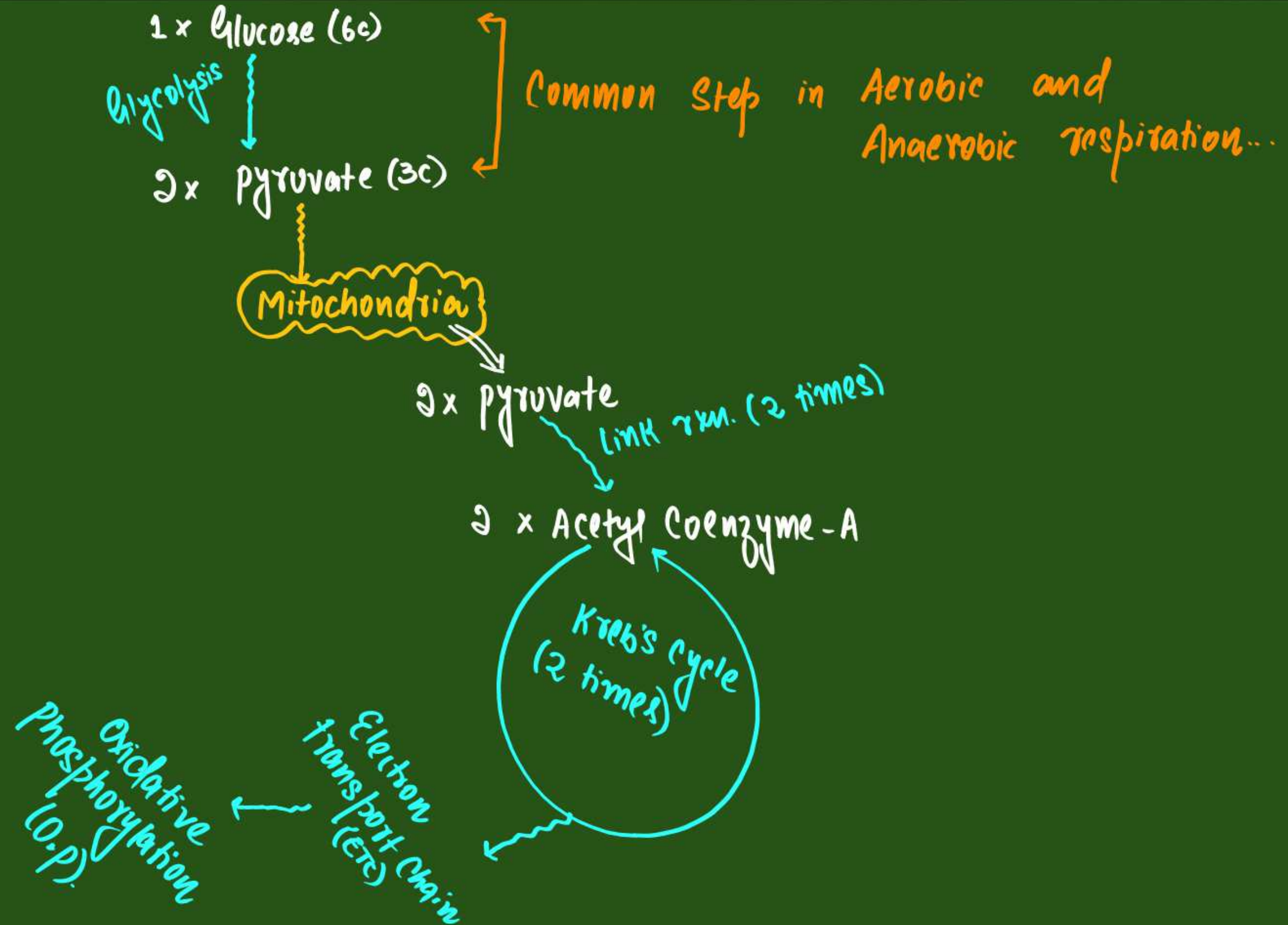
Here pyruvate undergoes complete oxidation to form:

i) 6CO_2

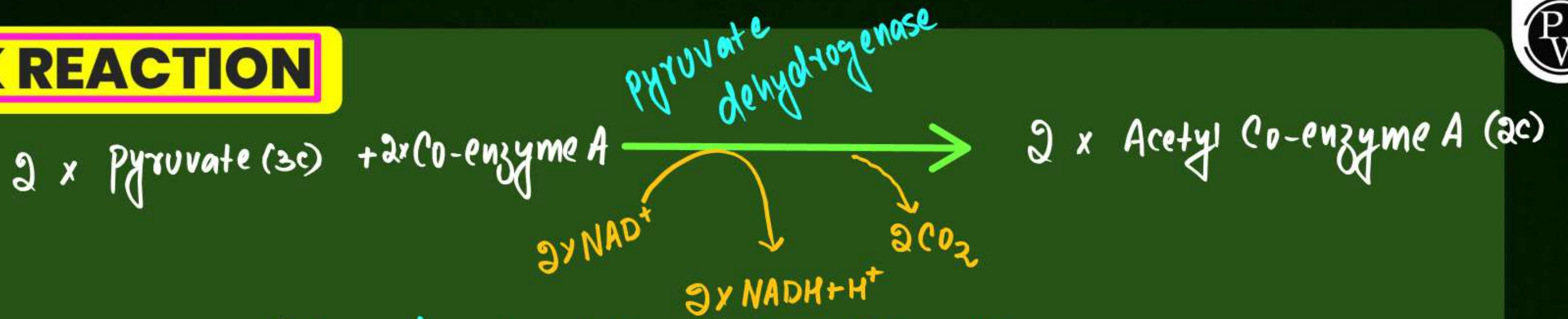


ii) large amount
of energy in the
form of ATP.





LINK REACTION



- Link rxn. is oxidative decarboxylation of pyruvate.
- Site: Mitochondrial matrix.
- Acetyl Co-A $\rightarrow$ Enters into "Kreb's Cycle"
- Thus Acetyl Co-A is a connecting link b/w Glycolysis and Kreb's Cycle.
- Link rxn is also called as transition rxn or Gateway rxn.

≡ from 1x Glucose $\rightsquigarrow$ 2x pyruvate are formed.

≡ 2x pyruvate $\rightsquigarrow$ 2x link rxns. occur (Thus for 01x Glucose, 02 link rxns occur).

From 02 link rxns.

$\rightarrow$ 2x CO_2

$\rightarrow$ 2 (NADH + H^+)

$\rightsquigarrow$ 2 (Acetyl CO-A)
(2c)

KREB'S CYCLE



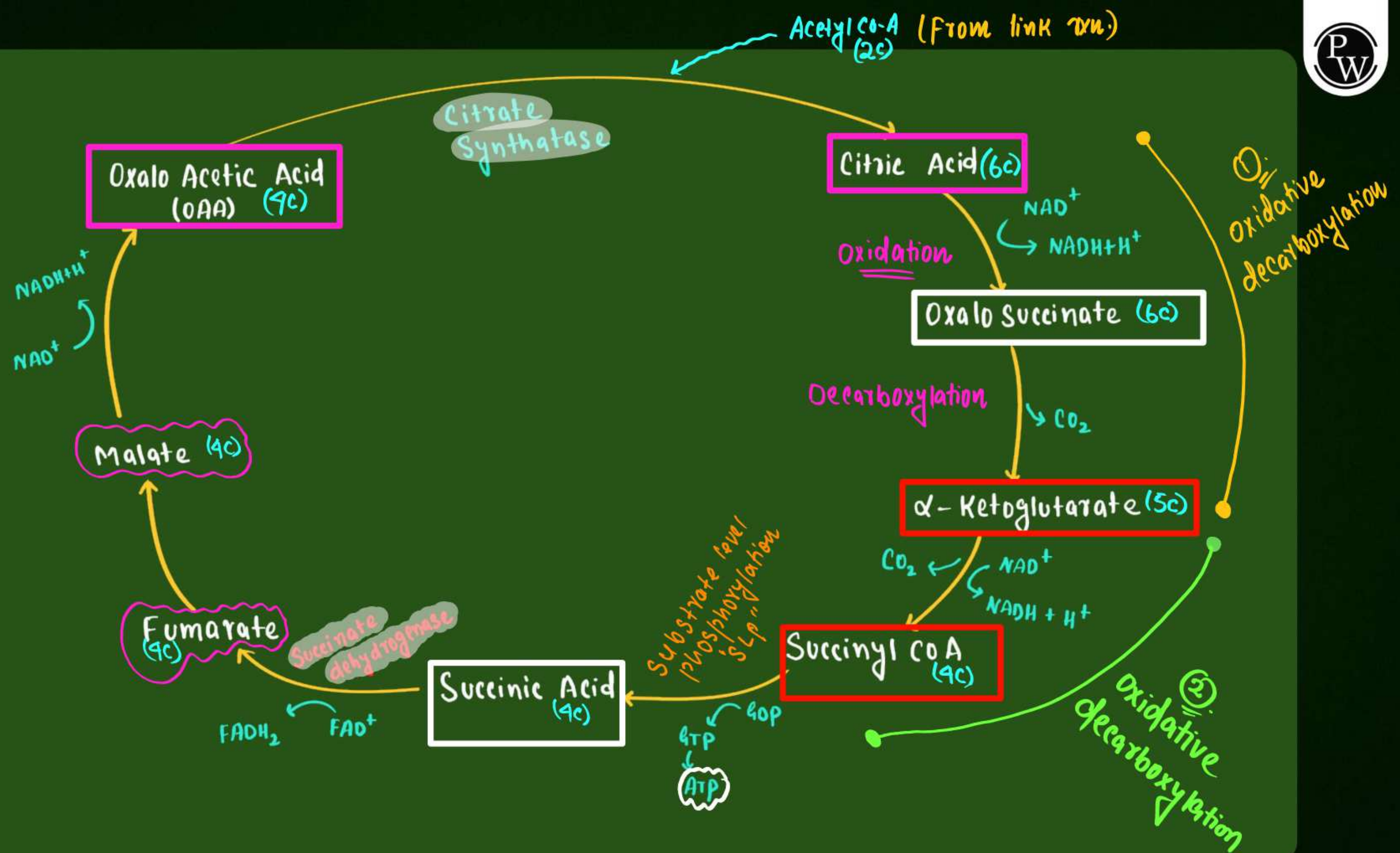
• Site: Mitochondrial matrix.

• Metabolic pathway of Kreb's cycle was worked out by Sir Han's Kreb i.e., why the name Kreb's cycle.

• First product formed is "Citric Acid"

• It consists of 03 Carboxyl (COO^-) groups.
i.e., why it is also called as TCA cycle.

↓
"Tri Carboxylic Acid cycle"



For Single turn of Kreb's Cycle

- No. of oxidative decarboxylation rxns. = 02.
- No. of CO_2 produced = 02
- ATP produced (SLP) = 01
- $\text{NADH} + \text{H}^+$ produced = 03
- FADH_2 produced = 01

For Single molecule of $\text{C}_6\text{H}_{12}\text{O}_6$, two turns of Kreb's Cycle take place

↓
For 02 turns of
Kreb's Cycle...

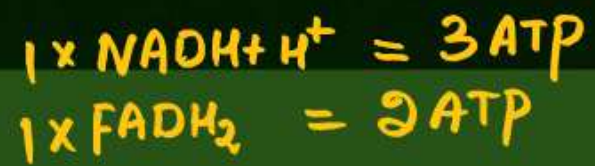
- No. of oxidative decarboxylation rxns. = 04
- No. of CO_2 produced = 04.
- ATP produced (SLP) = 02
- $\text{NADH} + \text{H}^+$ produced = 06
- FADH_2 produced = 02.

NOTE

All the enzymes of Kreb's cycle are present inside mitochondrial matrix except "Succinate dehydrogenase"

↓
located on inner mitochondrial membrane towards matrix side.

ATP BALANCE SHEET



	ATP (SLP)	$\text{NADH} + \text{H}^+$	FADH_2
$1 \times \text{C}_6\text{H}_{12}\text{O}_6$ $\downarrow$ glycolysis $2 \times \text{pyruvate}$	2 ATP	$3 \text{ NADH} + \text{H}^+$ $\downarrow$ o.p 6 ATP	X
• In aerobic respiration, glycolysis produces total 8 ATP • In anaerobic respiration, glycolysis produces total 2 ATP.			
$2 \times \text{pyruvate}$ $\downarrow$ Link rxn. (2 times) $2 \times \text{Acetyl Co-A}$	X	$3 \times \text{NADH} + \text{H}^+$ $\downarrow$ o.p 6 ATP	X
$2 \times \text{Acetyl Co-A}$ $\downarrow$ Krebs's cycle (2 times)	2 ATP	$6 \times \text{NADH} + \text{H}^+$ $\downarrow$ o.p 18 ATP	$2 \times \text{FADH}_2$ $\downarrow$ o.p 4 ATP

Total ATP produced from aerobic respiration of $1 \times C_6H_{12}O_6$ molecule is

38 ATP

Glycolysis *single turn*

- 2 ATP (SLP)
- 2 $NADH + H^+$ = 6 ATP
- total ATP = 8

Link rxn. *2 times*

- 2 $NADH + H^+$ = 6 ATP
- total ATP = 6 ATP

Kreb's Cycle *2 times*

- 2 ATP (SLP)
- 6 $NADH + H^+$ = 18 ATP
- 2 $FADH_2$ = 4 ATP
- Total ATP = 2 + 18 + 4 = 24

$$\begin{aligned} \text{Total ATP} &= 8 + 6 + 24 \\ &= 38 \end{aligned}$$

QUESTIONS

The 5-C compound formed during TCA cycle is

(NEET-2022 (Phase-2))

- A** Fumaric acid
- B** α -ketoglutaric acid
- C** Oxalo succinic acid
- D** Succinic acid

QUESTIONS

HLW

First oxidative decarboxylation in aerobic respiration occurs during the conversion of

- A** Oxalosuccinic acid to iso citric acid
- B** Pyruvic acid to acetyl CoA
- C** α -ketoglutaric acid to succinyl CoA
- D** Fumaric acid to malic acid

QUESTIONS

Which of the following enzyme of Krebs cycle is found attached to inner mitochondrial membrane?

- A** Succinate dehydrogenase
- B** Malic dehydrogenase
- C** Isocitrate dehydrogenase
- D** Citrate synthetase

HOMEWORK

1 Question
+
Revision....



Thank
You